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Saveetha Institute of Medical And Technical Sciences

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INTELLIGENT PNEUMONIA DETECTION AND DIAGNOSIS FROM CHEST X-RAY IMAGES USING DEEP LEARNING

Course Code:MLA0408

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ABSTRACT

- Develops an AI system to detect pneumonia from chest X-ray images.
- Preprocesses X-ray images for better model performance.
- Uses deep learning models to classify Normal and Pneumonia cases.
- Applies Explainable AI (Grad-CAM) for visual interpretation.
- Displays prediction results with confidence scores through a user-friendly interface.
- Improves diagnosis accuracy, speed, and clinical decision-making.

INTRODUCTION

- Pneumonia is a serious lung infection that requires early and accurate diagnosis to improve patient outcomes.
- Chest X-ray imaging is the most commonly used method for detecting pneumonia in clinical practice.
- Deep learning models can automatically analyze X-ray images and accurately identify pneumonia.
- Explainable AI (XAI) improves the transparency of model predictions, helping healthcare professionals trust the results.
- This project develops an AI-powered clinical decision support system for fast, reliable, and automated pneumonia detection.

PROBLEM IDENTIFICATION

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PROBLEM STATEMENT:

- Manual interpretation of chest X-ray images is time-consuming and depends on the availability and expertise of radiologists, which can lead to delayed diagnosis.
- Existing diagnostic methods may be prone to human error and are less effective in handling large volumes of medical images, highlighting the need for an accurate AI-based automated detection system.

CAUSES OF THE PROBLEM:

- The increasing number of pneumonia cases places a heavy workload on radiologists, making timely diagnosis more difficult.
- Variations in X-ray image quality and the similarity between normal and pneumonia-affected lungs can lead to inaccurate or delayed diagnosis.

MAIN OBJECTIVES

- To develop a deep learning-based system for automatic pneumonia detection from chest X-ray images.
- To improve the accuracy and speed of pneumonia diagnosis using Convolutional Neural Networks (CNN).
- To reduce human errors in the interpretation of chest X-ray images.
- To provide explainable AI-based predictions that support clinical decision-making.
- To design a user-friendly system for efficient and reliable pneumonia diagnosis.

LITERATURE SURVEY

Year	Author(s)	Method Used	Key Findings
2023	Zhang et al.	CNN	Achieved high accuracy in pneumonia detection from chest X-ray images.
2023	Kumar et al.	EfficientNet	Improved classification accuracy with fewer model parameters.
2024	Li et al.	ResNet + Grad-CAM	Enhanced prediction accuracy and provided visual explanations for diagnosis.
2024	Ahmed et al.	CNN with Data Augmentation	Increased model robustness and reduced overfitting.
2025	Sharma et al.	Hybrid CNN + Attention	Improved feature extraction and faster pneumonia detection.
2025	Patel et al.	Explainable AI (XAI)	Supported radiologists with reliable and interpretable AI-based diagnosis.

EXISTING SYSTEM AND PROPOSED SYSTEM

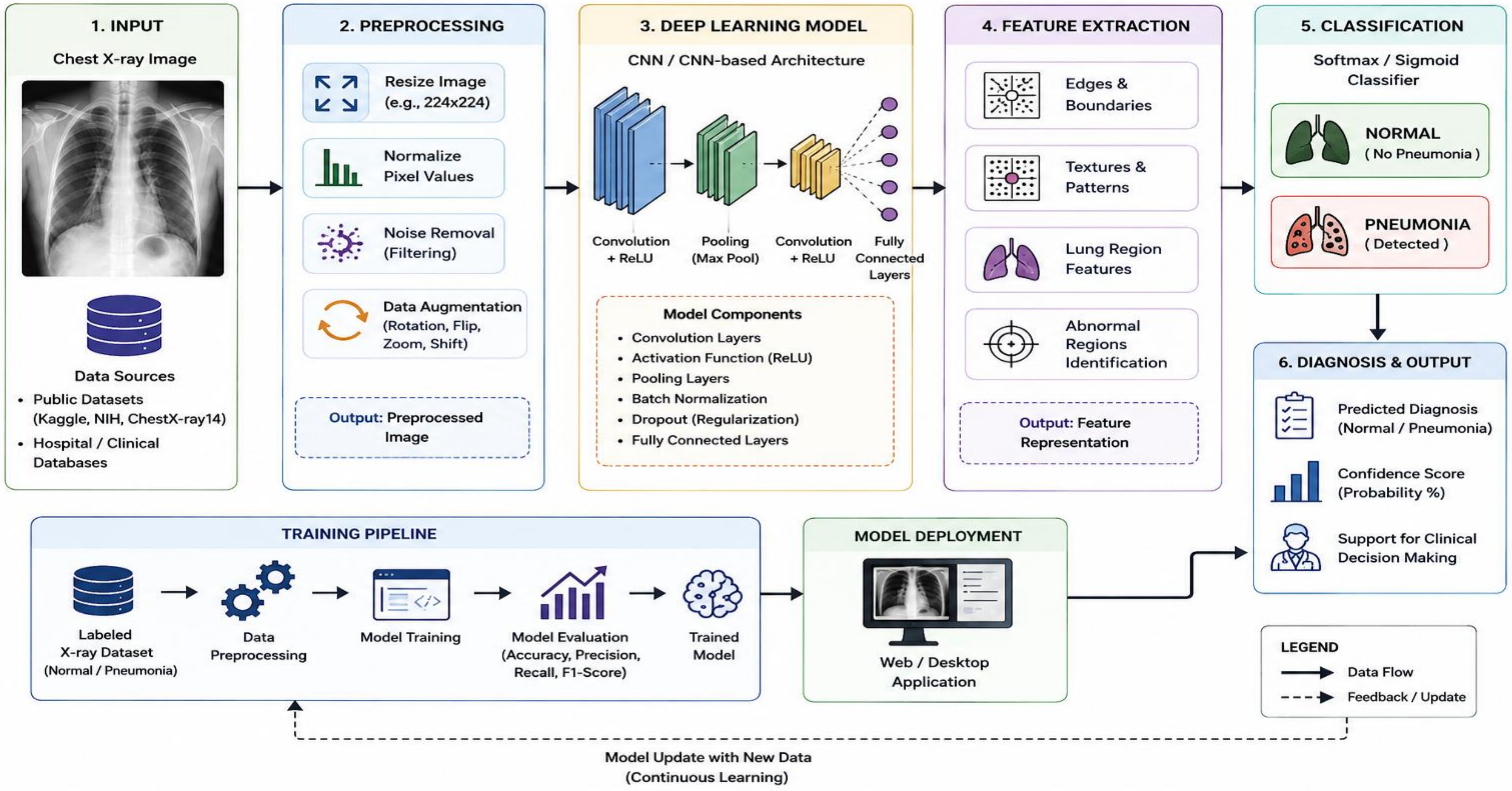
Existing System

- Manual interpretation of chest X-ray images by radiologists.
- Diagnosis is time-consuming and depends on expert availability.
- Prone to human errors and inconsistent results.
- Difficult to handle large numbers of X-ray images efficiently.
- Limited support for explaining diagnostic decisions.

Proposed System

- Uses deep learning (CNN/EfficientNet/ResNet) for automatic pneumonia detection.
- Preprocesses chest X-ray images for improved accuracy.
- Classifies images as **Normal** or **Pneumonia** automatically.
- Applies Explainable AI (Grad-CAM) to highlight affected lung regions.
- Provides fast, accurate, and reliable clinical decision support through a user-friendly interface.

SYSTEM ARCHITECTURE



Module 1: Data Collection and Image Preprocessing

•Collect chest X-ray images from publicly available medical datasets.

•Organize the dataset into training, validation, and testing sets.

•Resize and normalize images to a standard input size.

•Apply data augmentation techniques such as rotation, flipping, and zooming.

•Prepare the processed images for deep learning model training.

Fig-1 Input

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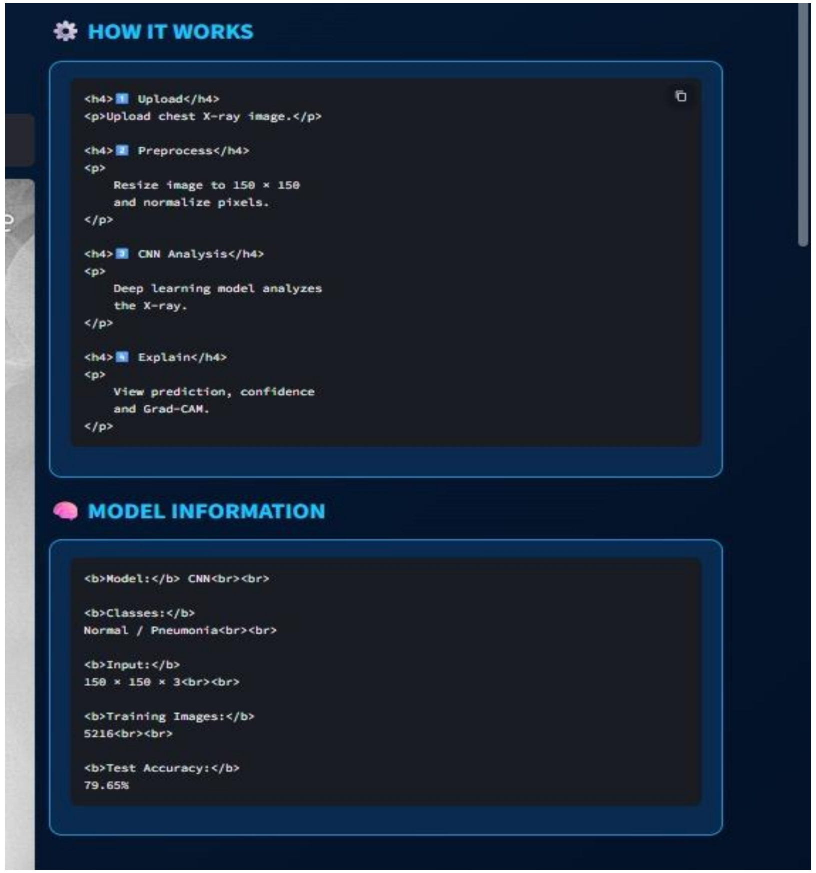
Module 2: Deep Learning Model Development

- Select a suitable deep learning model (CNN, EfficientNet, or ResNet).
- Train the model using the preprocessed chest X-ray images.
- Extract important image features automatically using deep learning.
- Classify images into Normal or Pneumonia categories.
- Optimize the model to improve prediction accuracy and reduce overfitting.



Module 3: Explainable AI and Prediction

- Accept a chest X-ray image as input from the user.
- Predict whether the image indicates Normal lungs or Pneumonia.
- Display the prediction with a confidence score.
- Use Explainable AI (Grad-CAM) to highlight the affected lung regions.
- Generate an interpretable report to support clinical decision-making.

Result

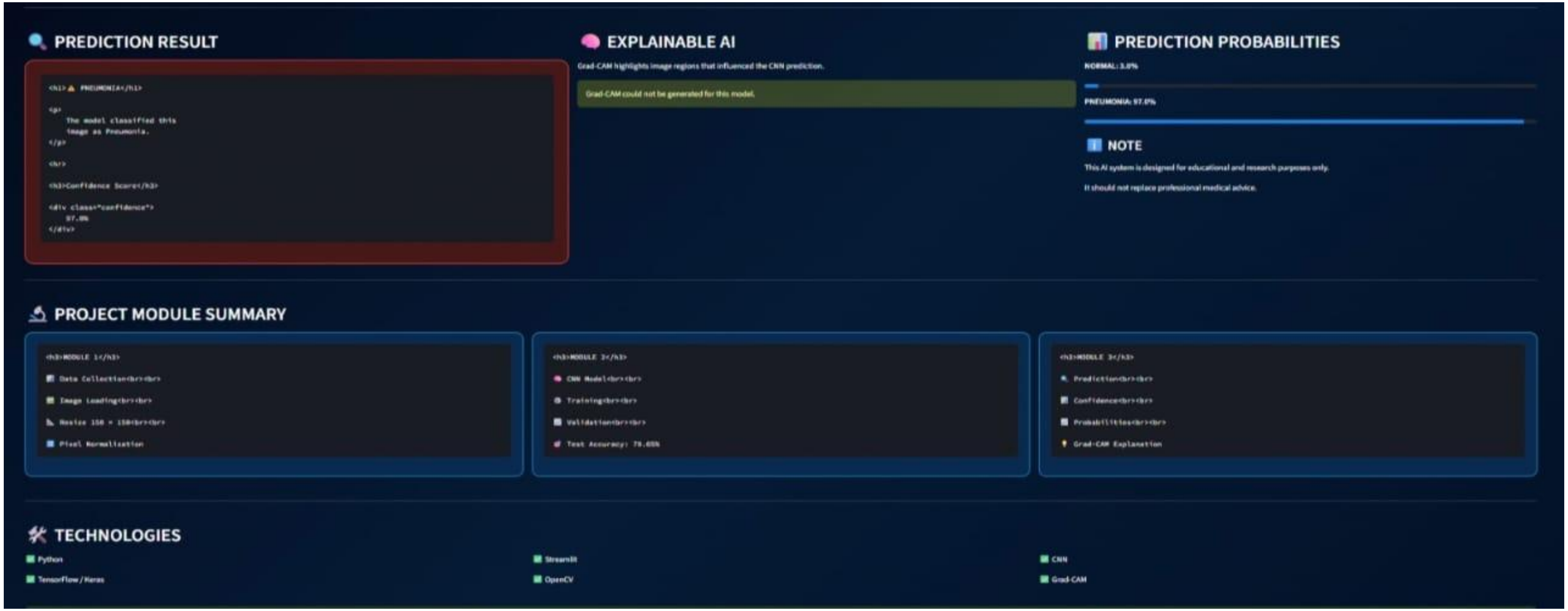

PREDICTION RESULT
<h1>⚠ PNEUMONIA</h1>
<p>
The model classified this
image as Pneumonia.
</p>
<hr>
<h3>Confidence Score</h3>
<div class="confidence">
97.0%
</div>
EXPLAINABLE AI
Grad-CAM highlights image regions that influenced the CNN prediction.
Grad-CAM could not be generated for this model.
PREDICTION PROBABILITIES
NORMAL: 3.0%
PNEUMONIA: 97.0%
NOTE
This AI system is designed for educational and research purposes only.
It should not replace professional medical advice.
PROJECT MODULE SUMMARY
<h3>MODULE 1</h3>
Data Collection

Image Loading

Pixel Normalization
<h3>MODULE 2</h3>
CNN Model

Training

Validation

<h3>MODULE 3</h3>
Prediction

Confidence

Probabilities

Grad-CAM Explanation
TECHNOLOGIES
Python
TensorFlow / Keras
Streamlit
OpenCV
CNN
Grad-CAM

Hardware / Software Required

Hardware

- Processor: Intel Core i5 or above
- RAM: 8 GB or more
- Storage: 256 GB SSD or above
- GPU: NVIDIA GPU (Optional)

Software

- Windows 10/11 or Ubuntu
- Python 3.10+
- Jupyter Notebook / VS Code
- TensorFlow / Keras
- OpenCV
- NumPy, Pandas, Matplotlib
- Scikit-learn

CONCLUSION

- Developed an AI-based system for automatic pneumonia detection.
- Used deep learning to improve diagnosis accuracy.
- Applied Explainable AI (Grad-CAM) for transparent predictions.
- Reduced diagnosis time and human errors.
- Supports healthcare professionals in making faster and more reliable clinical decisions.

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THANK YOU